3 Mount a keyless chuck in the tailstock to hold the drill bits needed to bore out the component. A little tip I picked up from one of Richard Findley's demonstrations is to give the chuck a light tap with a dead blow mallet, to seat it firmly in the Morse taper. This will help prevent the chuck from revolving in the taper when drilling or withdrawing the bit

4 Measure with callipers to determine the drill size required for the tip of the refill. As you can see in the photograph, mine measured just over 2.5mm, so a 2.5mm bit would be too small and 3mm, too loose a fit. Not having a full set of small metric drill bits in 0.1mm increments, I opted for a 7/64in imperial bit, which converts to 2.778mm, giving a nice sliding fit

5 Another important measurement is for the threads of the mechanism. Not all mechanisms are equal or alike. The one on the left in the photograph is taken from a cheap Chinese copy of a Sierra and is a comparatively coarse 8.5 x 1mm thread, while the one on the right from Beaufort is 8 x 0.75mm and compatible with original Sierra kits from quality suppliers. The drill and tap sizes you need will depend upon the mechanism you choose

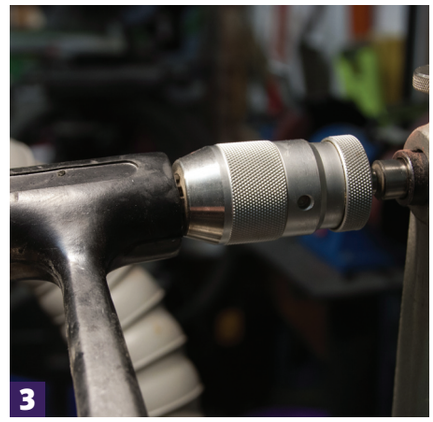
6 A standard 7/64in drill bit is not long enough to drill all the way through a 65mm blank. Therefore, drill this out first from what will become the tip end of the blank and then turn around the the blank and complete the remainder of the drilling from the other end. In order for this to be successful, the drilling must be accurate, so take your time and use good quality drill bits

7 Make a final check to ensure the fit is good by inserting the end of the refill in the bore. It should slide in easily, but be held more or less horizontally when released. If the fit is tight the refill will not retract against the spring pressure when released; if the refill flops about in the bore it will be difficult and annoying to write with

8 With the blank reversed in the chuck, now bore out to 6mm for the body of the refill. The bore needs to be deep enough to allow room for the compressed spring. Start at about 72mm and then gradually increase the depth until the correct fit is achieved

9 At this point it may be necessary to take the component off the lathe a few times to check the fit. The tip should freely extend by about 4–5mm when the spring is fully compressed. It will not be extended this far by the mechanism but this measurement will prevent over compression of the spring

10 Then open the bore out for about 10mm depth to the correct size, for whatever tap you will be using. Tap the hole using the methods described in my previous articles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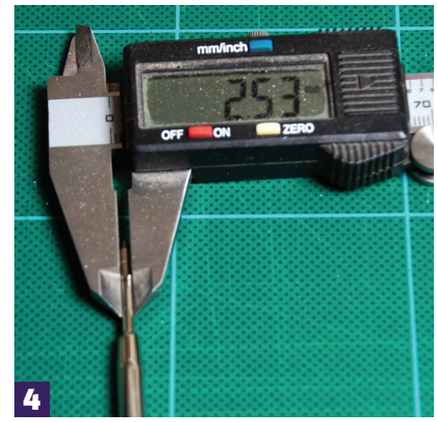

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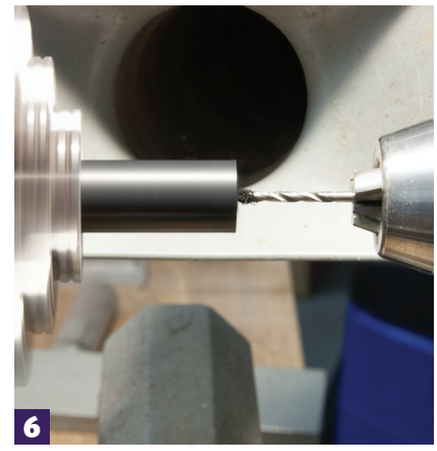
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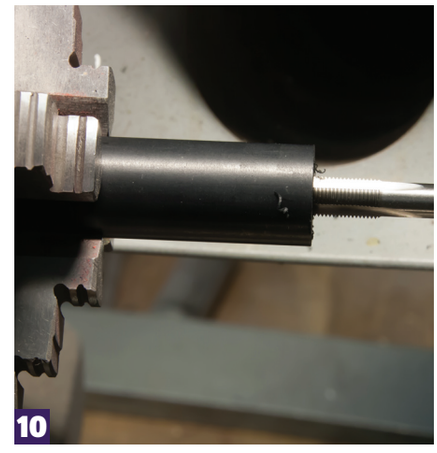
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